Storage Lab: Hybrid Cloud Storage Migration & RPO/RTO Data Protection Plan

Duration : 90 Min.

Scenario

An enterprise is migrating 500 TB of multi-tiered on-premises data (Fibre Channel SAN LUNs, NetApp NAS CIFS/NFS shares, and tape backup archives) to a hybrid multi-cloud environment leveraging AWS, Azure, and GCP. The migration architect must define the target storage service mapping, design the continuous replication and cutover strategy, establish disaster recovery RPO/RTO SLAs, configure cloud storage lifecycle policies (hot to cool to immutable archive), and design Azure Arc monitoring for remaining on-prem storage assets.

Task Objectives

Perform the following hybrid cloud storage migration and architecture blueprint tasks:

> **Pre-Staged Files in `student_workspace/`:** - **`migration_inventory.json`**: Source inventory of 500 TB multi-tier enterprise storage. - **`hybrid_storage_migration_blueprint.json`**: Starter architecture specification with `TODO_*` placeholders.

1. Review `migration_inventory.json` and map on-premises SAN block LUNs, NAS file shares, and tape archives to cloud equivalents.
2. Formulate quantifiable RPO and RTO disaster recovery SLA targets across mission-critical, operational, and archive tiers.
3. Define automated time-based lifecycle transition rules from high-performance tiers to immutable cold archives.
4. Produce the complete architecture specification in `hybrid_storage_migration_blueprint.json`.

Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.
2. Execute the required storage management commands, volume operations, or configuration edits.
3. If configuration files, backup manifests, or diagnostic reports are required, save them inside the `student_workspace` directory.
4. Verify your storage configurations locally by executing: `bash python run.py`
5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Block workload mapping: comprehensive IOPS, throughput, and capacity sizing for SAN-to-cloud block migration	3 Marks
TC2	File share mapping: protocol parity (NFSv3/v4, SMB3) and permission preservation (NTFS ACLs, POSIX) defined	3 Marks
TC3	Archive & compliance mapping: immutable WORM retention and lifecycle policies formulated across cloud tiers	3 Marks
TC4	Lifecycle transition rules: precise time-based tiering rules from hot to cool to archive configured	2 Marks
TC5	RPO/RTO SLA framework: quantifiable RPO and RTO targets defined with justification per business tier	3 Marks
TC6	Cutover & delta sync plan: phased synchronization methodology minimizing production cutover maintenance window	2 Marks
TC7	Azure Arc hybrid monitoring: architectural design incorporating Azure Arc for unified on-prem/cloud visibility	2 Marks
TC8	Blueprint validation: hybrid_storage_migration_blueprint.json conforms to standard architecture schema	2 Marks

Total Score: 20 Marks

Important Notes

- This is an **Auto-Evaluated** storage assessment.
- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.